

En un espacio de Hilbert, convergencia débil y en norma implica convergencia fuerte

Proposición 1. Sea H un espacio de Hilbert real y $(x_n)_{n \in \mathbb{N}} \subset H$, entonces

$$x_n \rightharpoonup x \wedge \|x_n\| \rightarrow \|x\| \implies x_n \rightarrow x \text{ en } H.$$

Demostración: Basta ver que $\|x_n - x\| \rightarrow 0$:

$$\begin{aligned} \|x_n - x\|^2 &= \langle x_n - x, x_n - x \rangle = \|x_n\|^2 - 2 \langle x_n, x \rangle + \|x\|^2 \\ &= \|x_n\|^2 - 2 (L_x(x_n)) + \|x\|^2. \end{aligned}$$

Al tomar límites cuando $n \rightarrow \infty$ en la expresión anterior, obtenemos que

$$\|x_n - x\|^2 \xrightarrow{n \rightarrow \infty} \|x\|^2 - 2 (L_x(x)) + \|x\|^2 = \|x\|^2 - 2 \|x\|^2 + \|x\|^2 = 0.$$

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